

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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firm regardless of AI exposure. For workers aged 22-25, we find a 12 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially.

Sixth, the above facts are largely consistent across various alternative sample constructions. We find that our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022, around the time of rapid proliferation of generative AI tools.⁴ They also hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we caution that the facts we document may in part be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that, by nature of the model training process, AI replaces codified knowledge, the “book-learning” that forms the core of formal education. AI may be less capable of replacing

⁴OpenAI introduced ChatGPT in November 2022.

4.5 Fact 5: Labor market adjustments are visible in employment more than compensation

In addition to employment we observe workers’ annual base compensation. We use this information to test for labor market adjustment along the compensation margin.¹⁹ Salary data are deflated to 2017 dollars using the PCE index.²⁰

Results are in Figure 10. The findings indicate a less marked divergence in compensation compared to employment across more and less exposed occupations. Figure 11 shows results by age and Eloundou et al. (2024)-based exposure quintile. We find little difference in compensation trends by age or exposure quintile.

Prior work by Autor and Thompson (2025) notes that technology that replaces inexpert tasks may reduce occupational employment but increase occupational wages; technology that replaces expert tasks may do the opposite. The sign of the wage effect depends on the overall share of tasks displaced as well as the whether these tasks are expert or inexpert. The limited changes we find for wages suggest that these effects may be offsetting, at least in the short run. Alternatively, the results could be explained by wage stickiness in the short run, consistent with recent evidence from Davis and Krolikowski (2025).

4.6 Fact 6: Findings are largely consistent under alternative sample constructions

We test the robustness of these results to alternative sample constructions and robustness checks.

Excluding Technology Occupations One possibility is that our results are explained by a general slowdown in technology hiring from 2022 to 2023 as firms recovered from the COVID-19 Pandemic.²¹ Figure A4 shows employment changes by age and exposure quintile after excluding computer occupations, corresponding to 2010 SOC codes that start with 15-1. Figure A5 shows results when excluding firms in the information sector (NAICS code 51). Results are quite similar,

¹⁹Total compensation may additionally include bonuses, overtime pay, commissions, equity, tips, and other items. These may have a greater impact on overall compensation in certain professions and age groups than others.

²⁰In contrast to the series for employment, results for compensation end in June 2025, the most recently available month for the PCE index.

²¹Under the Tax Cuts and Jobs Act, amendments to Internal Revenue Code §174 enacted in 2022 also disallowed companies from immediately deducting R&D expenditures, including software development costs. These costs instead had to be capitalized and amortized over five years for domestic research and fifteen years for foreign research.

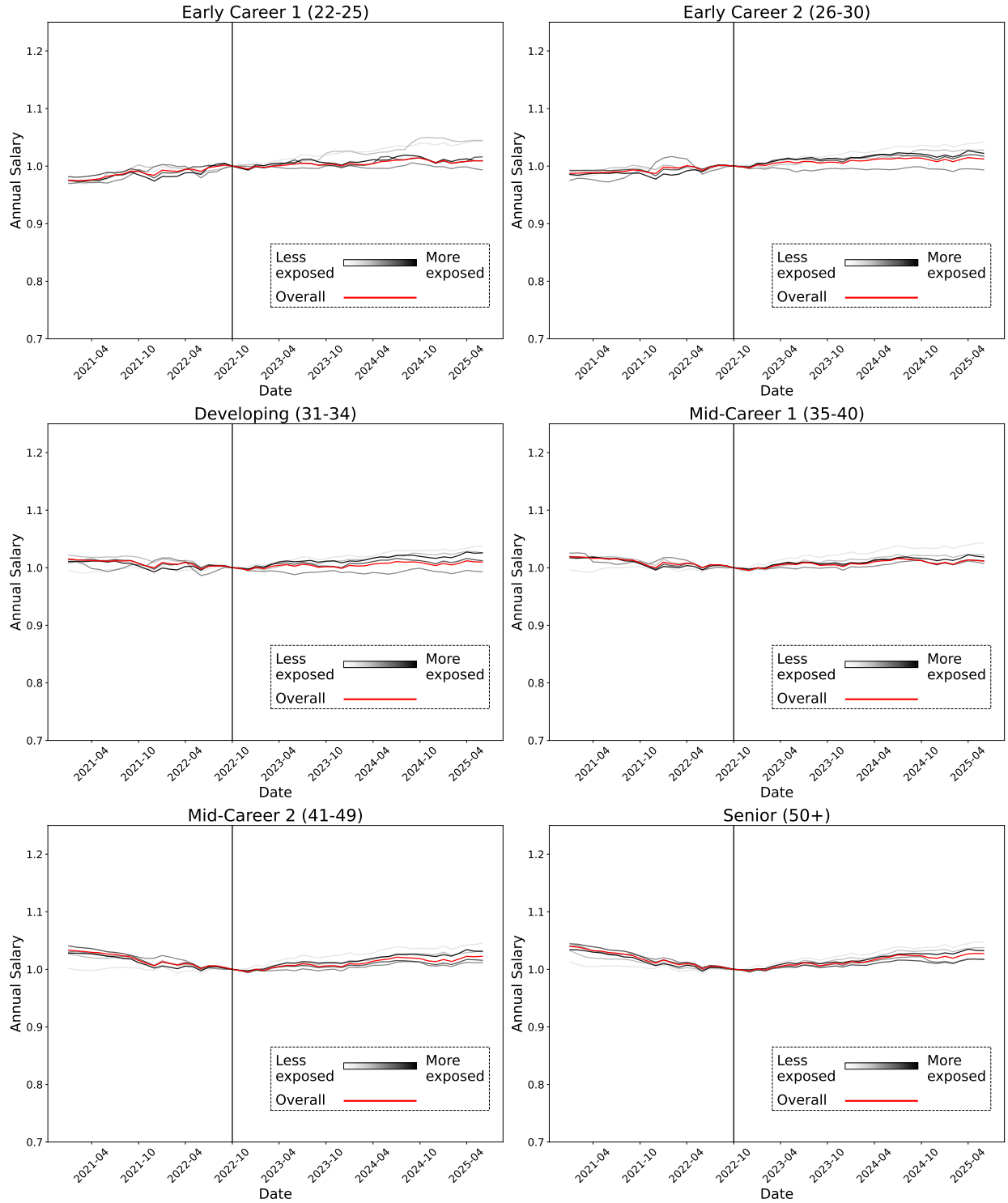


Figure 11: *Annual base compensation*. Changes in annual base compensation by age and exposure quintile. Exposure quintiles are defined based on the GPT-4 β measures from [Eloundou et al. \(2024\)](#). Darker lines are more exposed quintiles. The red line shows the overall trend pooling across quintiles. Annual base compensation is deflated to 2017 dollars using the PCE deflator.